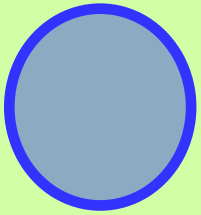
Enzymology
Second stage
October 2026



Introduction

Enzymes are extraordinary efficient, they are biocatalysts, or biological polymers or organic catalysts in biologic systems. Catalysts are substances that alter the speed of chemical reactions without themselves undergoing any consumption and permanent changes.

Enzymes are proteins in nature, soluble in water and can be precipitated by protein precipitating agents such as: trichloroacetic acid, tungestic acid, alkaline zinc salts, organic solvents ---------etc.

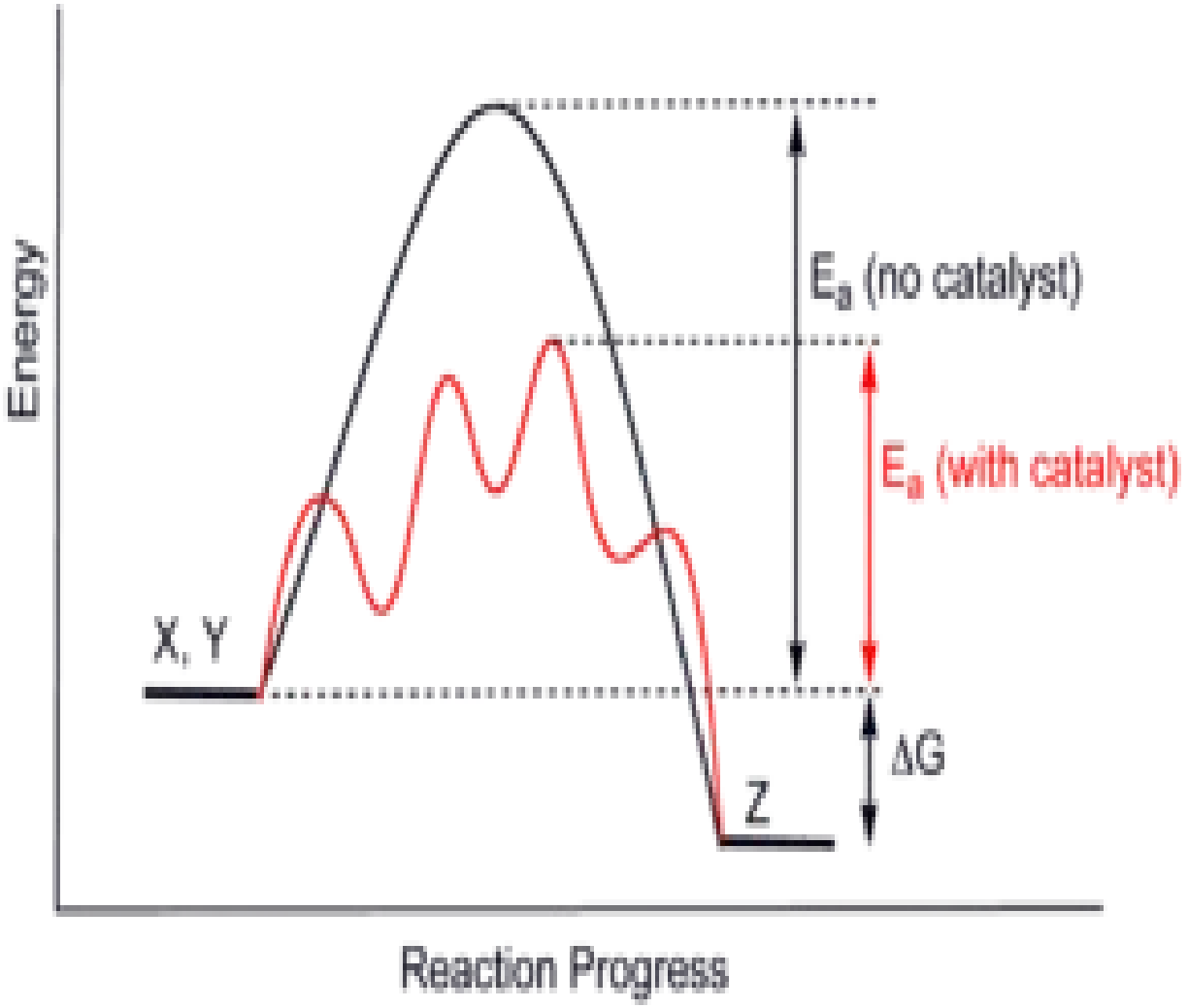
Enzymes increase the rate of reaction by changing the way a reaction takes place, but is itself not change at the end of reaction

Increase the rate of reaction r= Δ [P]/Δ t

But not the equilibrium of the reaction.

Enzymes like all catalysts has lower activation energy.

(rate-limiting free energy of activation)than the corresponding unanalyzed reaction.


Energy
E_a (no catalyst)
E_a (with catalyst)
X, Y
ΔG
Z
Reaction Progress

?What do enzymes do

- Enzymes catalyse the many chemical reactions which together make up the process of metabolism
- An enzyme speeds up the rate at which a reaction reaches equilibrium, e.g.

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

- For this reaction:
  - un-catalysed rate is 220 molecules per hour
  - catalysed rate is 600,000 molecules per second
  - catalysed by the enzyme *carbonic anhydrase*
- Enzymes DO NOT affect the equilibrium position of a reaction

Enzymes [E], at the first step react with substrate [S] produce complex [ES], the second step [ES] complex convert to product [P], and [E]. So the substrate convert to product, and the enzyme remain as the same without consume or any change in structure, cause increase rate of chemical reaction and decrease activation energy of the reaction.



:Enzymes are characterized by great

Affinity, activity, specificity and susceptibility to the influence of pH ,temp. and other environmental changes.

Enzymes catalyzed reactions are typically 10^3-10^{20} **times faster than the corresponding uncatalyzed reactions.**

Like proteins, enzymes are more stable in concentrated solutions, and they are subjected to denaturation by changes in pH, temp. of their solutions, because of which the enzyme activity may be lost .

Depending on their chemical composition, enzymes are classified into:

1-Simple protein enzymes: they consist of only specific type of protein, such as lysozymes (catalyse the hydrolysis of bacterial cell wall), found for example in tears and behaves as an antibiotic.

2-Complex protein enzymes: they contain in addition to specific type of protein additional substances

(prosthetic group) such as co-enzymes and metals .

CO-ENZYMES (enzyme helpers = teeth of enzymes)

These are specific non protein organic part of complex protein enzymes which are essential for enzyme activity on substrate(any substance acted upon by an enzyme).

The complete system of enzyme is called Holo-enzyme :

Holo-enzyme

- Protein part (apo-enzyme), heat unstable, high molecular weight
- Non protein part (co-factor): co-enzyme and metals, heat stable low molecular weight .

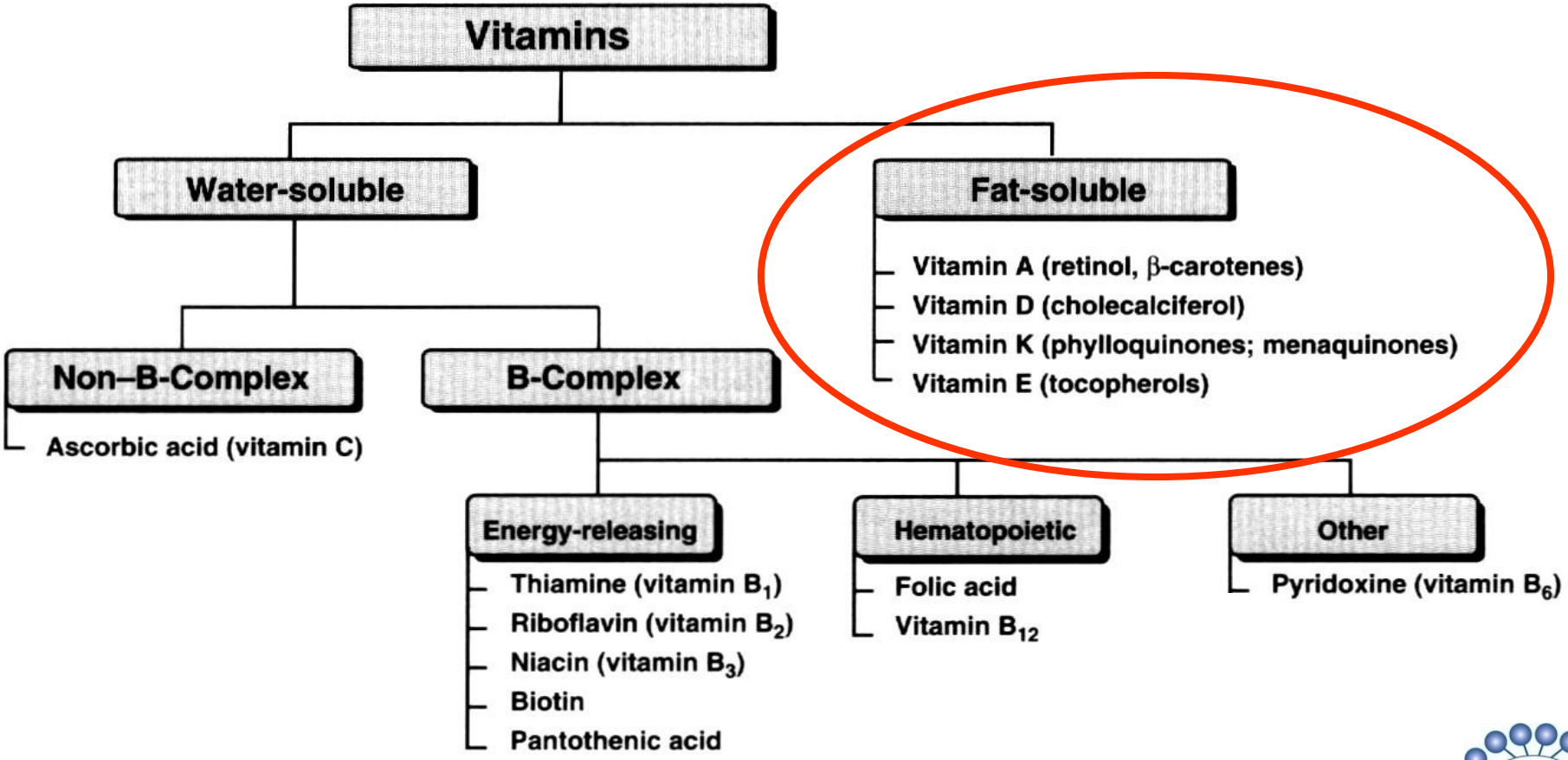
Co-enzymes generally are derivatives of water-soluble vitamins which include vitamin B complex {thiamin B1, riboflavin B2, niacin B3, pyridoxine B6, Folic acid (Tetra hydroflavin TH4) .B9, cobalamin B12}, and vitamin C (ascorbic acid)

Vitamins



VITAMINS are organic compounds, the human body can not be synthesized, for that must be obtain with the diet. Vitamins are micronutrients, small amounts needs (less than 100 mg/day). **Vitamins** are required for normal growth, development and maintenance of health.

Vitamins: The Family Tree

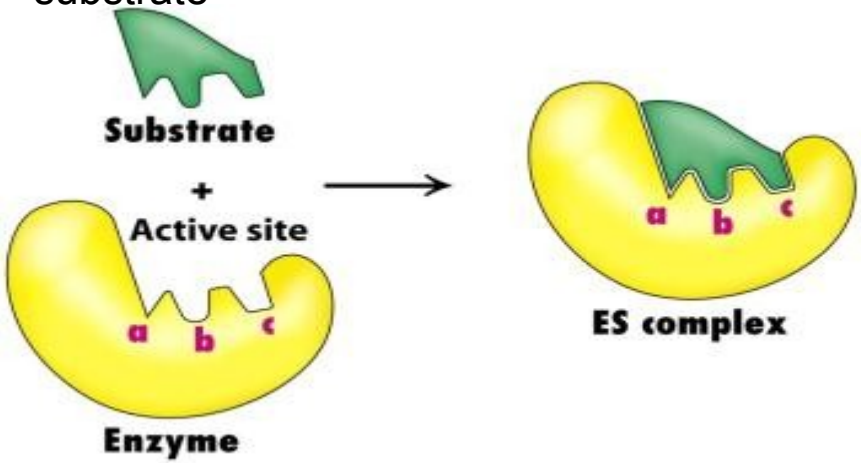


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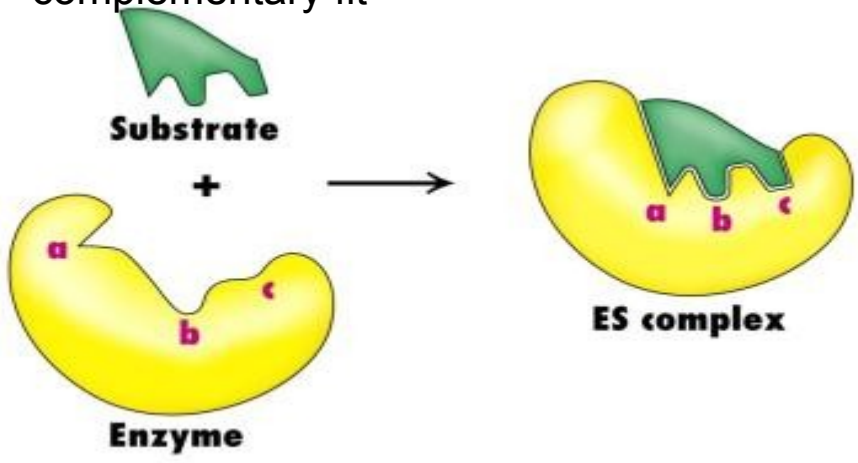
Substrate binding to enzyme

- Substrate binds to an active site
  - a cleft or crevice
  - contains amino acids essential for catalytic activity
  - contains amino acids for highly specific interactions

Lock-and-key model
Active site of unbound enzyme is complementary to the shape of the substrate



Induced fit model
Binding of substrate induces a conformational change in enzyme, results in complementary fit



;Active site

Is three-dimensional cavity of the enzyme with specific chemical properties that enable it to accommodate the substrate.

FACTORS AFFECTING ENZYME ACTIVITY

A few factors influence enzyme activity;

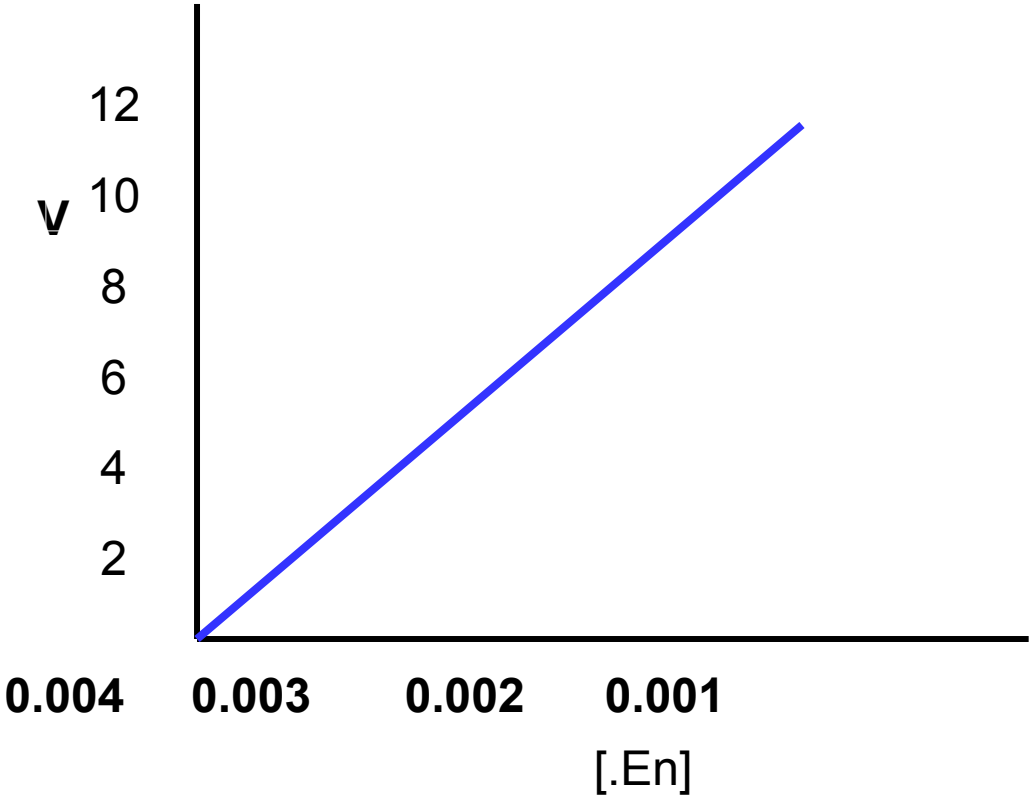
the principals are the followings :

1-Enzyme concentration { En }:

Within wide limits the speed of an enzymatic reaction is proportional to the enzyme concentration when the substrate concentration is maintained constant, as its illustrated by the following figure:

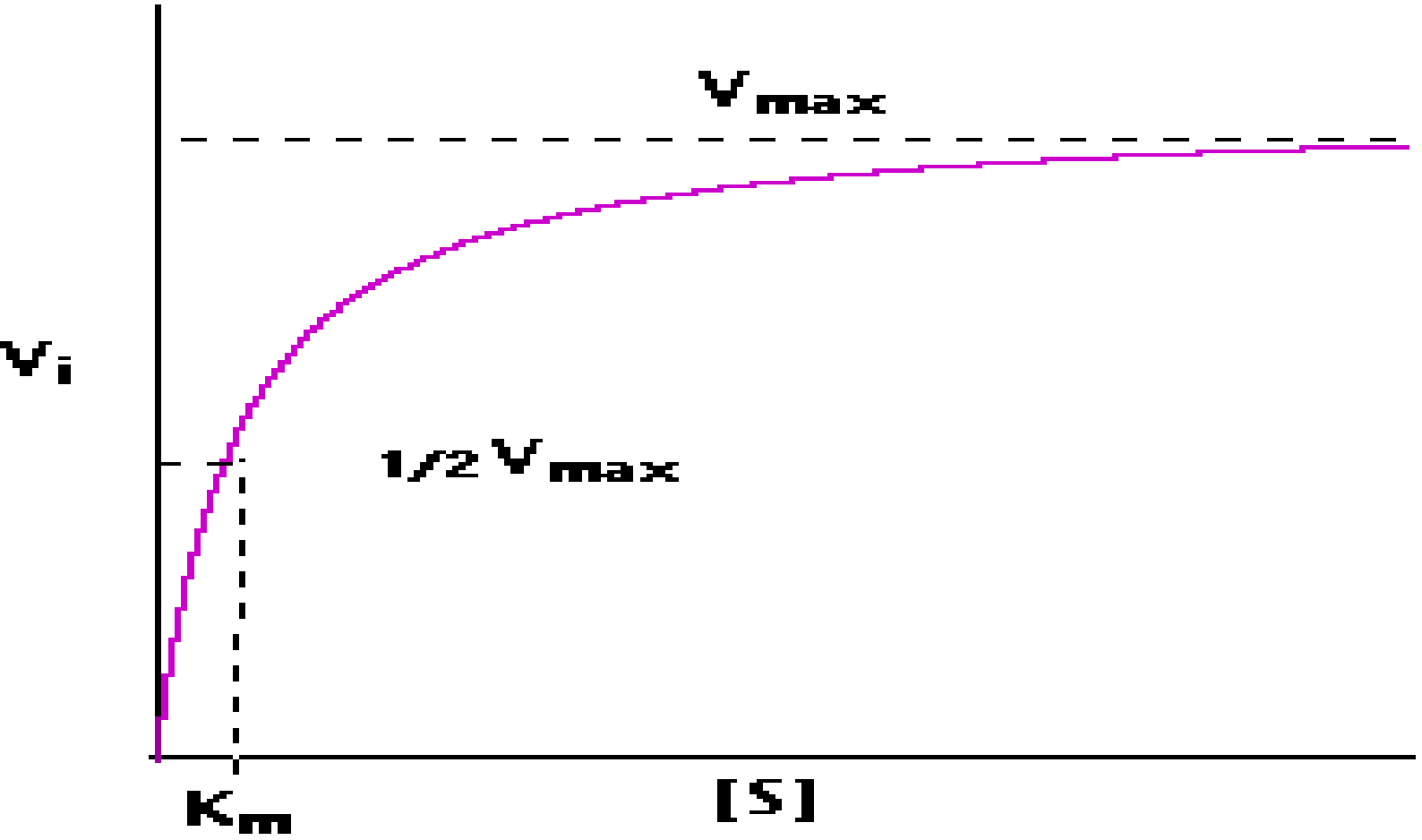
$V \propto \{ En \}$

$V = K \{ En \}$



2-Substrate concentration [S]:

If the concentration of the substrate [s] is increased while all other conditions are kept constant, the velocity of an enzyme catalysed reaction increases to maximum value and then no further increase, since the enzyme will be saturated with substrate.



V_i = initial velocity (moles/time)

[S] = substrate concentration (molar)

V_{max} = maximum velocity

K_m = substrate concentration when V_i is one-half V_{max} (Michaelis-Menton constant)

A +B $\longrightarrow$ P

Rate α [A] [B]

Rate α [P]

Keq is a ratio of rate center

K1 [A] [B] =k-1[P]

K1/K-1=[P]/[A][B]

The Michales Menten equation

The effects of substrate concentrations;

$$v_o = \frac{[V_{max}][S]}{Km + [S]}$$

v_o = initial reaction velocity

V_{max} = maximal velocity

[S] = substrate concentration

The substrate concentration that produces a V_i that is one-half of V_{max} is designated the Michaelis-Menten constant, K_m (named after the scientists who developed the study of enzyme kinetics).

K_m is (roughly) an inverse measure of the affinity or strength of binding between the enzyme and its substrate. The lower the K_m, the greater the affinity (so the lower the concentration of substrate needed to achieve a given rate).

- The physiological significance of Km is illustrated by the sensitivity of some individuals to **alcohol;** such persons exhibit facial flushing and tachycardia (increased heart rate) after intake of even small amount of alcohol.
- In the liver ethanol (alcohol) is converted to acetaldehyde by an enzyme alcohol dehydrogenase. Normally the cause of symptoms of alcoholic is due to the accumulation of acetaldehyde in the body.
- Acetaldehyde which is then oxidized to acetate by liver enzyme acetaldehyde dehydrogenase.
- Most people have two forms of the enzyme acetaldehyde dehydrogenase.

-a low Km mitochondrial form and

-a high Km cytoplasmic form.

In susceptible persons, the mitochondrial enzyme is less active due to the substitution of a single amino acid and acetaldehyde is converted to acetate only by cytoplasmic enzyme. Because this enzyme has a high Km (less affinity to substrate) less acetaldehyde is converted to acetate and leads to accumulation of acetaldehyde in blood which accounts for the physiological effects of alcoholics.

Hypersensitivity to alcohol exists when drinking small amounts of alcohol causes **facial flushing and tachycardia (rapid heartbeat).** Alcohol dehydrogenase generates acetaldehyde, which is converted to acetate by aldehyde dehydrogenase. The latter enzyme exists in two forms, a high-affinity (low Km) form and a low-affinity (high Km) form. Those sensitive to alcohol lack the high-affinity form, **resulting in excess acetaldehyde and, hence, vasodilation.**

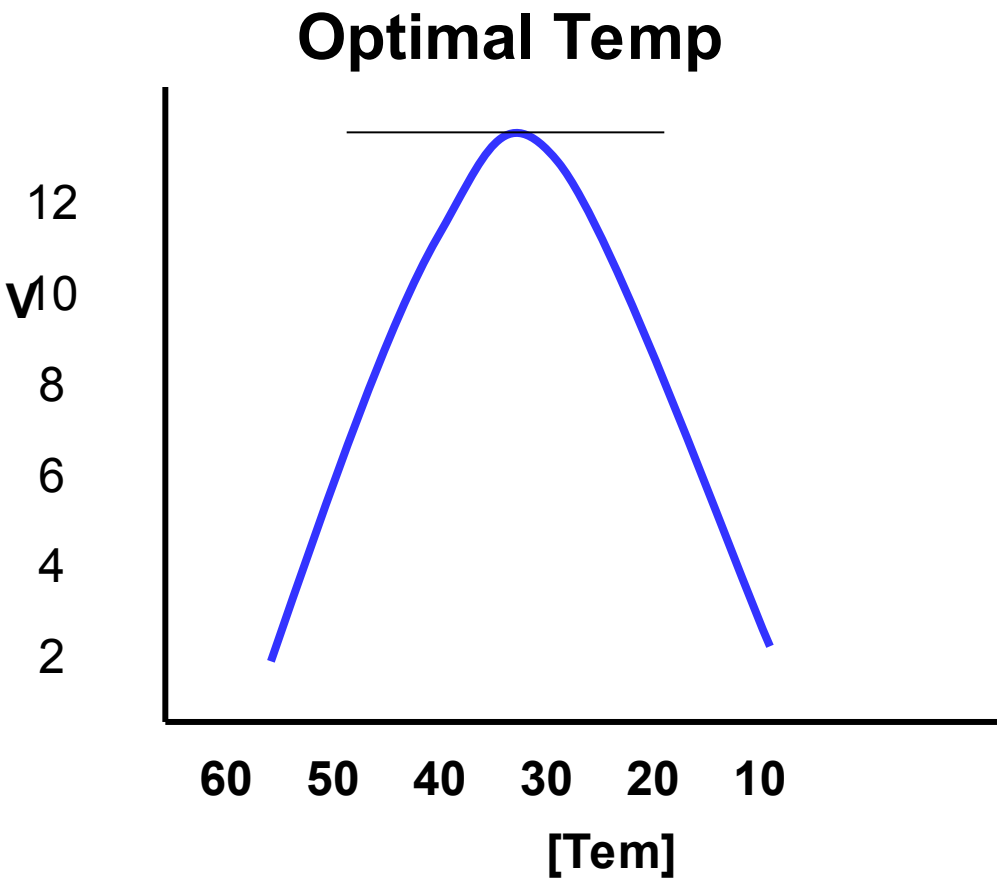
3-Temperature:

The optimum temperature of an enzyme is that temp. at which the greatest amount of substrate is changed in unit time. For example, in the case of G.I.T. enzymes, the optimum temp. is around 40 C.

Generally, the velocity of an enzyme catalyzed reaction increases as temp. rises.

For example, the velocity of many biochemical reactions double with a ten-degree centigrade rise in temp. and this is known as temp. coefficient or Q10.

$$Q10 = \frac{\text{Reaction rate at } T + 10}{\text{Reaction rate at } T}$$


Optimal Temp
12
V10
8
6
4
2
60
50
40
30
20
10
[Tem]

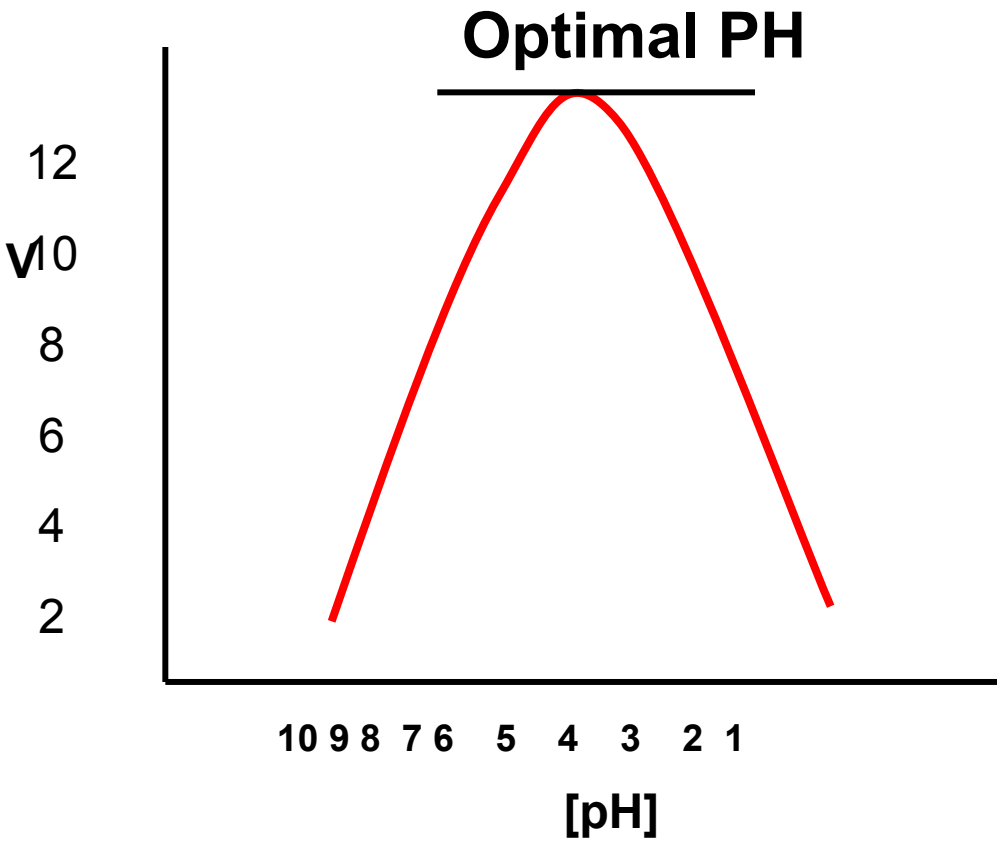
4-pH :

Each enzyme has an optimum pH at which the enzyme reacts at maximum speed or has the greatest activity.

Generally moderate pH changes affect the ionic states of enzyme and substrate .

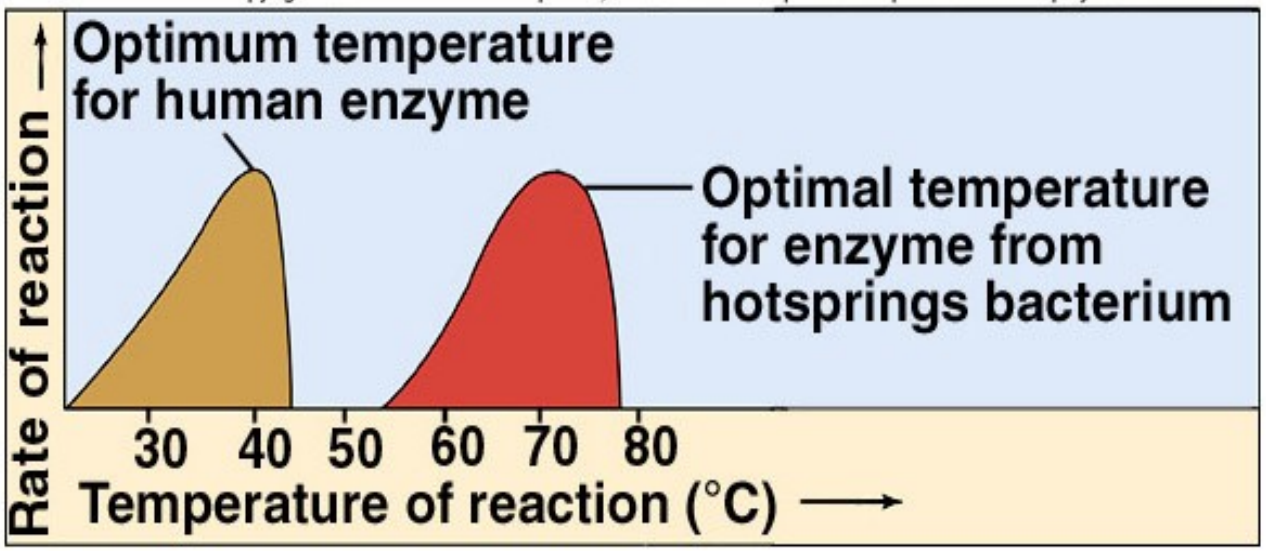
Most enzymes are inactivated in strong acid or alkaline solutions, for example pepsin loses its activity rapidly above pH 8 and sucrase below pH 3.

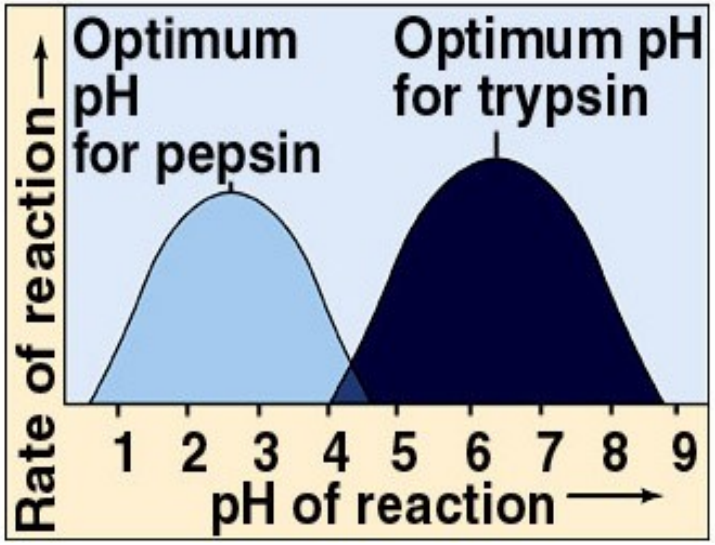
Shortly the optimum pH of most enzymes is between pH5 and pH 8.


Optimal PH
12
V 10
8
6
4
2
10 9 8 7 6 5 4 3 2 1
[pH]

Temperature and pH affect enzyme activity

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Relationship of Enzyme Activity to Environment

5-Concentration of reaction products = end-product inhibition = feed-back inhibition = retro-inhibition :

It's known that the addition of products of an enzyme catalysed reaction to a system contains purified enzyme destroys the usual relationship between enzyme concentration and the reaction rate, for example:

En 1 En 2 En 3

A ----------B ----------C ----------D (end product)

In this example the end product (D) has more affinity to En 1 than that of A so D combines with En 1 forming En D complex, consequently the reaction rate is inhibited.

Sucrase or Invertase

Sucrose ⟶ Glucose + Fructose

The end-products (Glucose and Fructose) have also affinity for sucrase forming complexes and thus inhibiting the hydrolysis of sucrose .

Enzymes that show end-product inhibition are known to have at least two binding sites (active site and allosteric site) .

6-Physical agents :

Enzyme activity may be lost by U.V light, X-ray and even by strong shaking of their solutions .

7-Metal activators:

Many ions and molecules have the capacity to activate certain enzymes, for example many enzymes need a metal ion for activity and in some cases the requirement is specific for a particular metal, for example: carbonic anhydrase shows no activity upon removing of Zn and no other metal is known to replace Zn in this enzyme. Over 25% of all enzymes contain tightly bond metal ions or require them for activity. These enzymes are called metallo-enzymes.

Metals	Enzymes
Cu	Tyrosinase, Phenolase, Ascorbic acid Oxidase
Fe	Catalase, Peroxidase, Tryptophan Oxidase, Cytochrome oxidase
Ca	Lipases, Proteases
Zn	Carbonic anhydrase, Alcohol Dehydrogenase , Lactic Dehydrogenase
Mn	Dipeptidase
Mg , Co	Peptidase, Phosphotransferase

8-Enzyme inhibitors or enzyme inhibition:

Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.

The pharmacological action of many drugs and pesticides depends largely on enzyme inhibition and in general there are three types of enzyme inhibition:

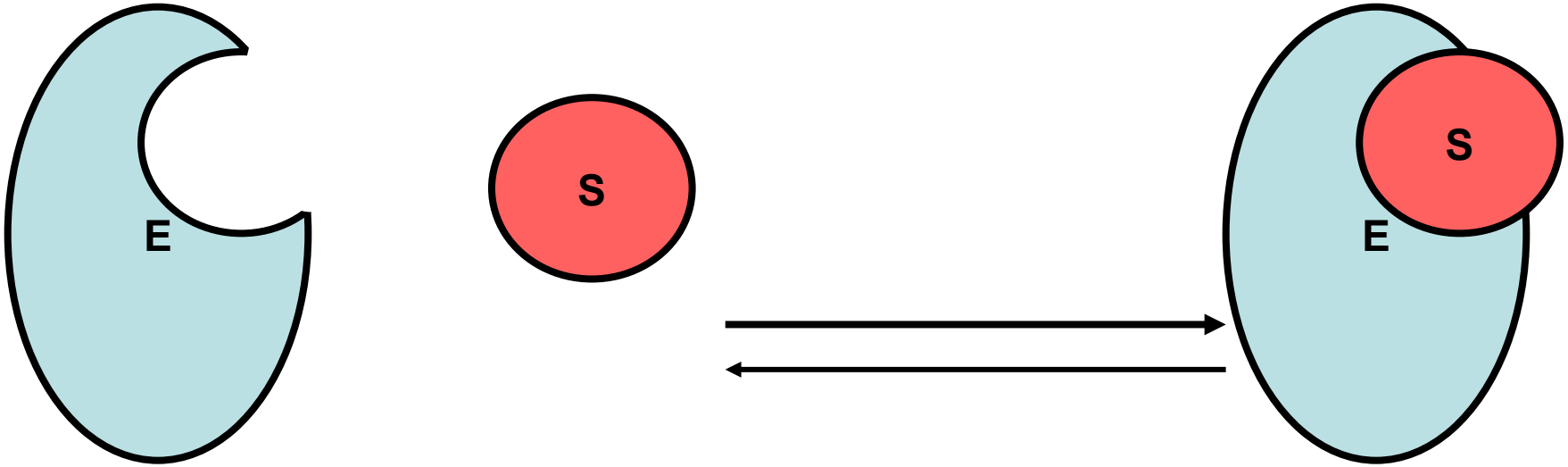
1-Competitive enzyme inhibition or substrate analogue inhibitor.

2-Noncompetitive enzyme inhibition.

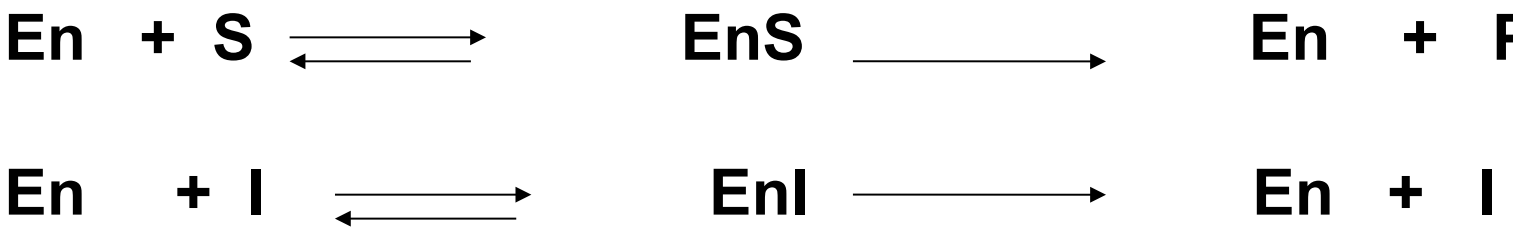
3-Un competitive enzyme inhibition.

1-Competitive enzyme inhibition:

This type of enzyme inhibition occurs at the substrate binding site:



The inhibitor resembles closely in structure to substrate so it can combine reversibly with the enzyme forming an EnI complex instead of EnS complex which is normally formed. When both the substrate and this type of inhibitor are present, they compete for the same binding site on the enzyme surface.



Competitive inhibition is usually reversible.

Excess substrate abolishes the inhibition.

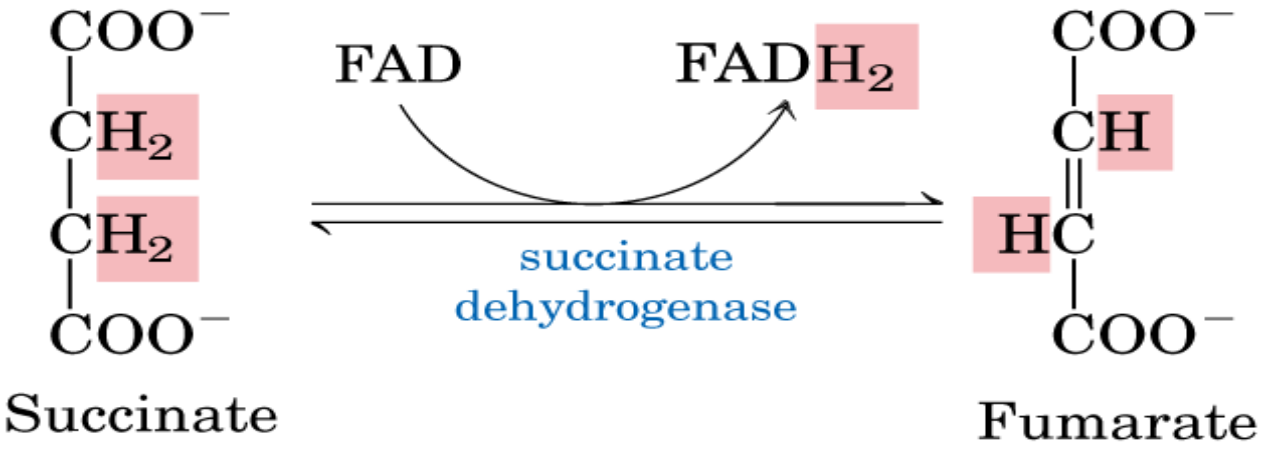
Examples

1-

Succinic acid = S

Malonic acd = I

Succinate dehydrogenase = En



$$\Delta G'^{\circ} = 0 \text{ kJ/mol}$$

Normally the enzyme succinate dehydrogenase removes two hydrogens from the substrate succinic acid to form fumaric acid. When malonic acid is present which has certain similarities in chemical structure, it competes with the substrate (succinic acid) for the enzyme succinate dehydrogenase, consequently the reaction is inhibited .

Malonic acid (COOH—CH2—COOH).

2-Competitive enzyme inhibitors that inhibit enzyme catalysed biochemical reactions in living systems are potent chemotherapeutic agents:

A-

PABA = S

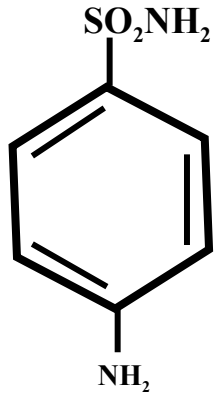
Sulfa drugs = I

Enzyme = Series of enzymes

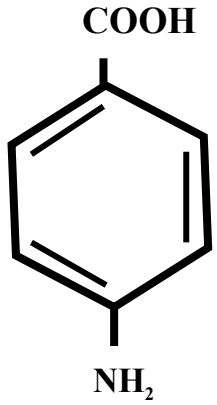
Series`of biochemical reactions Catalysed by multiple enzymes

PABA → → → → → → Folic acid

(an essential vitamin for growth and multiplication of Micro-organisms)



General structural formula of sulfa drug



PABA

In micro-organism in a series of biochemical reactions catalysed by multiple enzyme systems, PABA is converted to folic acid an essential vitamin for growth and multiplication of micro-organisms.

When sulfa drugs a structural analogue to PABA is present, the biosynthesis of folic acid is inhibited, and this is fatal for micro-organisms. Since man lacks the enzymes necessary for folic acid synthesis and folic acid must be present in the diet, so in man sulfa drugs do not act as a competitive inhibitor.

In addition, micro-organisms can not absorb folic acid from the host.

B-FH2 = S

Methotrexate = I, is a drug (chemotherapy) used for rheumatoid arthritis, leukaemia and other cancer.

Dihydrofolate reductase = En

$$FH2 \xrightarrow{\textit{Dihydrofolate reductase}} FH4$$

(FH4 an important co-enzyme in biosynthesis of purine and pyrimidine bases , so DNA which has an important role in cell division).

When methotrexate a structural analogue to FH2 is present, it enters in competition with FH2 for the enzyme dihydrofolate reductase, consequently FH4, purine and pyrimidine bases, DNA are not formed, and cell division will be stopped .

C- The family of drugs known as Angiotensin Converting Enzyme (ACE) inhibitors are a good example of enzyme inhibitors as drugs .



S ; inactive (decapeptide)

P ; potent pressor ,

Captopril, enalapril, and lisinopril = I

(octapeptide)

D-Non-Steroidal Anti-Inflammatory Drugs (NSAID)

Non-Steroidal Anti-Inflammatory Drugs (NSAID), such as aspirin, which inhibit the enzyme cyclo oxygenase, an important enzyme for conversion arachidonic acid to prostaglandins (mediate inflammation and pain).

E-Statins

Statins are group of drugs such as Atorvastatin, they inhibit production of cholesterol, take by patients with hypercholesterolemia, this drug inhibit the enzyme 3-hydroxyl-3methyl glutaryl CoA reductase (HMG CoA reductase), this enzyme catalyze convert HMG CoA to mevalonic acid, a rate-limiting step for cholesterol production, so cause decrease cholesterol in blood.

F-The treatment of CH_3OH intoxication using CH_3CH_2OH depends also on competitive enzyme inhibition

Alc.dehydrogenase *Aldehyde oxidase*

$CH_3OH \longrightarrow H\text{-}CHO \longrightarrow H\text{-}COOH$

Methanol Formaldehyde

Toxic to optic nerve

Alc.dehydrogenase Aldehyde oxidase

$CH_3CH_2OH \longrightarrow CH_3\text{-}CHO \longrightarrow CH_3COOH$

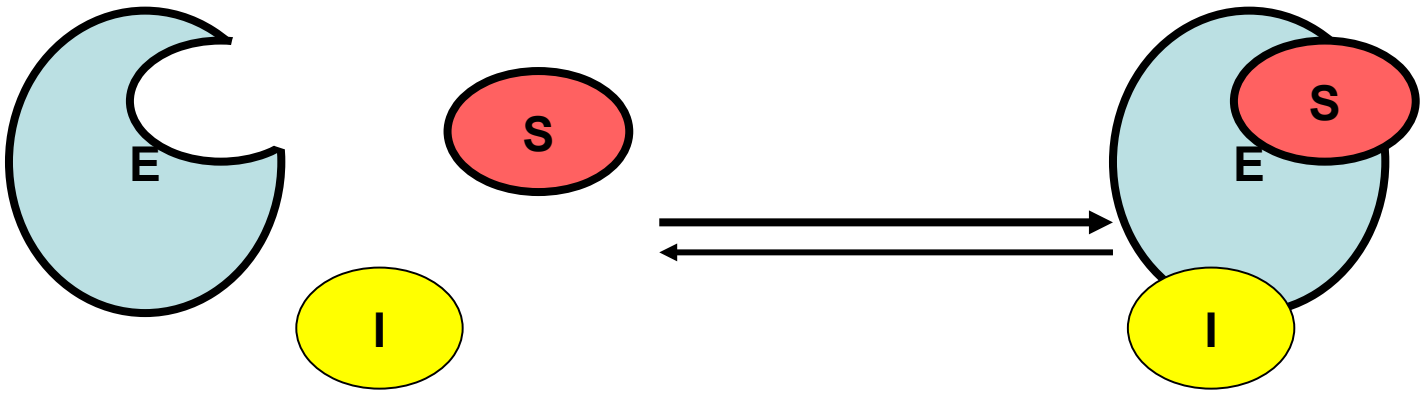
Ethanol Acetaldehyde

2-Non-competitive enzyme inhibition: this type of enzyme inhibition is divided into two subtypes:

A-Reversible non-competitive enzyme inhibition:

In this type of inhibition there is no structural similarities between the substrate and the inhibitor, so there is no competition between them for the same active site of the enzyme. So, the inhibitor is not bind to the same region of the enzyme that do the substrate, however the inhibitor can also form complex with the enzyme (EnI) through other points on the enzyme surface causing a conformational change of the active site




E
S
I
S
E
I

Example of reversible non-competitive inhibitors:

A-Trypsine inhibitors

The inhibitory of alpha-1-antitrypsine on proteases is reversible non-competitive type.

B-Antibody prepared against any enzyme will precipitate the enzyme, but the inhibition is also reversible.

b-Irreversible non-competitive enzyme inhibition:

Several enzyme poisons such as: iodoacetamide, oxidizing agents and heavy metals like Ag, Hg, Pb.



Nerve gases, agents of chemical terrorism such as tabun, sarin, and alkylating insecticides (malathion) are irreversible inhibitors of acetyl cholinesterase.

-Cyanide inhibits cytochrome oxidase

-Fluoride remove Mg and Mn ions and so will inhibit the enzyme enolase and consequently the glycolysis.

A common noncompetitive inhibitor is a result of chelation (**Metal binding**) therapy, such as **Ethylene diamine tetra acetic acid (EDTA),** resulting in removal of required divalent metal ions from the active site of enzymes. Blood of patients is collected in tubes with EDTA to inhibit both **calcium-activated proteases and the blood coagulation pathway.**

3-Uncometitive inhibition

Here inhibition does not have any affinity for the enzyme. Inhibitor binds to E-S complex, but not to the free enzyme.

Inhibition of placental ALP (Regan iso-enzyme) by phenyl alanine is an example of uncompetitive inhibition.

Suicide inhibition:

It is a special type of irreversible inhibition of enzyme activity; it is also known as mechanism-based inactivation.

Example;

Allopurinal is structurally like xanthine, so act as a competitive inhibitor for xanthine oxidase, and Allopurinal oxidize by the enzyme to Alloxanthine and act a strong inhibitor.

CLASSIFICATION AND NOMENCLATURE OF ENZYMES

Many enzymes have been named by adding the suffix (ase) to the name of substrate upon which the enzyme acts, thus enzymes act upon starch (Amylon) were termed Amylase, those act on lipids (Lipos) were termed Lipase and those act on protein, protease and so on.

However, this system of naming and classification of enzymes has not always been practical. For this reason and because the number of newly discovered enzymes is increasing rapidly, in 1961 the Enzyme Commission (EC) of the International Union of Biochemistry (I.U.B.) propose a systematic naming and classification of enzymes.

This new system divides biochemical reactions and enzymes catalyzed them into six principal classes and a number of subclasses (4—13) according to the reaction types. This new system is complex but more: precise, informative and descriptive

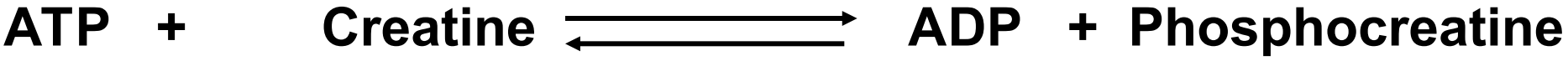
and according to this new system each enzyme has:

1-A recommended (Trivial) name which is short for daily use

2-A systematic name which identifies the reaction catalyzed.

3-A classification number consists of four elements separated by periods .

Example



Donor Acceptor

1-Recommended name = Creatine kinase

2-Systematic name = ATP : Creatine phosphotransferase

3- Classification number = 2.7.3.2

The first element of classification number (2) shows to which of the six principal classes this particular enzyme belongs . The six principal classes are:

1-Oxido-reductases . 2-Transferases ;

3-Hydrolases . 4-Lyases .

5-Isomerases . 6- Ligases or Synthetases

The second element of the classification number (7) shows the nature of the reaction catalysed by indicating for example the nature of the group transferred (phosphotransferase).

The third number of classification number(3) shows the nature of the acceptor substrate which is creatine. The fourth element of the classification number (2) shows the nature of the donor (ATP) which is a co-enzyme.

So, we can say that shortly the elements (1.2) indicate the class and the subclass of the enzyme, and the elements(3,4) indicate the nature of the substrates (donor and acceptor).

1-Oxidoreductases:

These are enzymes catalyse oxido-reduction reactions between substrates and subdivided into five subclasses.

Oxidases -1-1

$$AH_2 + 1/2O_2 \longrightarrow A + H_2O$$

Aerobic dehydrogenase-1-2

$$AH_2 + O_2 \longrightarrow A + H_2O_2$$

Anaerobic dehydrogenase-1-3

$$AH2 + B \longrightarrow A + BH_2$$

Hydroperoxidases-1-4

A-Peroxidase

$H_2O_2 + 2GSH \xrightarrow{\textit{Glutathione peroxidase}} 2H_2O + GSSG$

B-Catalase

$2H_2O_2 \xrightarrow{\textit{Catalase}} 2H_2O + O_2$

Oxygenases-1-5

A-Mono-oxygenases

$AH + O_2 + BH_2 \xrightarrow{\textit{Mono-oxygenase}} AOH + B + H_2O$

$A + O2 \xrightarrow{\textit{Dioxygenase}} AO2$

:Transferases-2

These are enzymes catalysing reactions involving transfer of group from molecule to other molecule, such as creatine kinase, glucokinase and others.

Creatine +ATP —*Kinase*→ **Creatine phosphate + ADP**

3-Hydrolases:

These are enzymes catalysing reactions involving hydrolysis, such as maltase, sucrase, protease.

Maltose + H_2O $\xrightarrow{Maltase}$ **Glucose + Glucose**

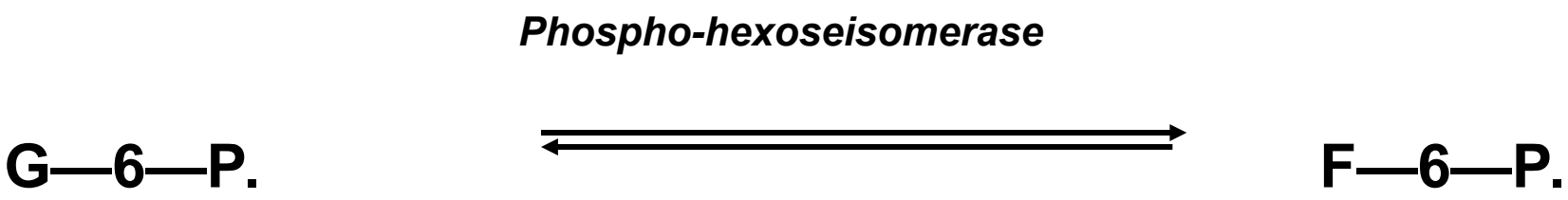
4-Lyases:

These are enzymes catalyse the removal of atoms or groups from substrates by mechanisms other than hydrolysis leaving double bond (s), such as fumarase.

Fumarate + H_2O —*Fumarase*→ **Malate**

5-Isomerases:

These are enzymes catalyzing interconversion of optical, positional and geometrical isomers, such as cis-trans isomerase, for example Phospho-hexoseisomerase.



6-Ligases or Synthetases :

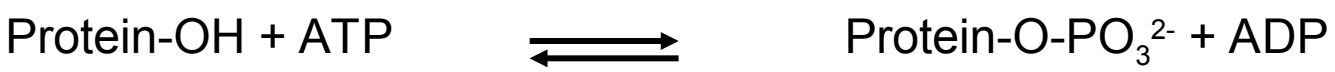
These are enzymes catalysing the linking together of two elements or compounds, depending on the chemical nature of the bond which is formed, example citrate synthase.

Oxaloacetate + Acetyl CoA $\xrightarrow{\textit{Citrate synthase}}$ Citrate

$$NH_3 + CO_2 + 2ATP \xrightarrow{\textit{Carbamoyl phosphate synthetase}} \text{Carbamoyl phosphate} + 2ADP + Pi$$

Regulation of enzyme activity by phosphorylation

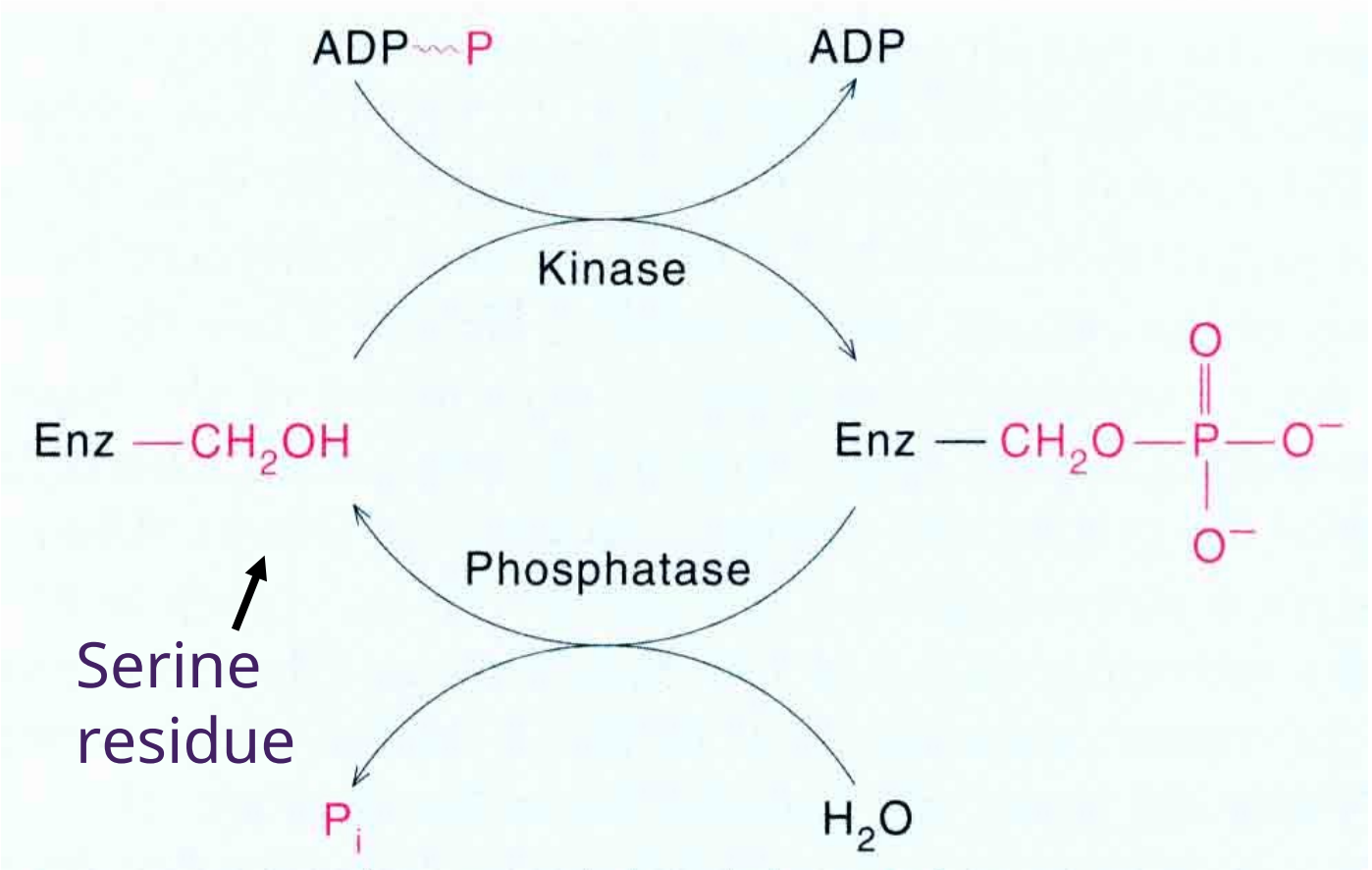
- An example of reversible covalent modification
  - side groups of serine, threonine and tyrosine can form phosphate esters



- Can convert enzyme to active or inactive form
  - e.g. glycogen phosphorylase
  - phosphorylation converts inactive a form to active b form

- Phosphorylation reactions are carried out by protein kinases
  - very fast, reversible and found in all cells, especially involved in energy generating processes

Kinases and phosphatases



Regulation of enzyme activity by irreversible covalent modification

- Result in activation of enzymes
- *Zymogens* – inactive precursors of an enzyme – are irreversibly transformed into active enzymes by cleavage of a covalent bond
  - in pancreas: trypsinogen and chymotrypsinogen, inactive precursors, are formed
  - in small intestine: enteropeptidase cleaves trypsinogen to form active trypsin which cleaves chymotrypsinogen to form active chymotrypsin

:Other examples –

- digestive enzymes, blood-clotting enzymes, clot-dissolving enzymes

ZYMOGENS AND ISO-ENZYMES (ISOZYMES)

Zymogens (pro-enzymes) : these are the inactive precursors of enzymes ,examples:

Zymogens	Active form of enzymes
Pepsinogen	Pepsin
Trypsinogen	Trypsin
Prothrombin	Thrombin
Proelastase	Elastase

CLINICAL CORRELATION

C-peptide links the insulin A and B chains in proinsulin. The peptide is removed by proteolytic processing via **carboxypeptidase E**, yielding active insulin. **Mutations in carboxypeptidase E** rendering it inactive or lowering its activity may cause **hyperproinsulinemia and diabetes in the homozygous state.**

Isoenzymes)) Isozymes

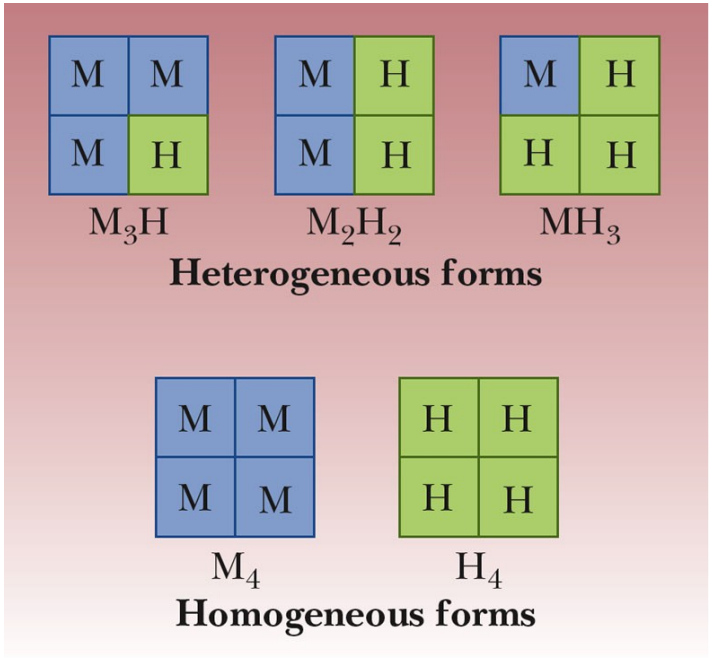
- Isozymes are isoforms of enzymes; they catalyse the same reaction but have different properties and structure (and sequence)
- Different isozymes can be
  - synthesised during different stages of foetal and embryonic development
  - present in different tissues
  - present in different cellular locations

Different isoenzymes often have different Km values for the same .substrate, ex. Quoted

Glucokinase	Hexokinase
,High Km (5-6 mmol) Low affinity	,Low Km (0.05 mmol) High affinity
Operates with the liver, since the liver monitors glucose and can both import and export glucose to maintain the blood glucose conc. About 5mmol/L (90 mg/dl)	Hexokinase located in muscle, where it is responsible for initiating glycolysis to provide energy for muscle conc. Muscle only imports and uses glucose, it does not export.
Activity depend on insulin	Activity not depend on insulin
Not inhibit by glucose-6-phosphate	Inhibit by glucose-6- phosphate

Isozymes – tissue specific isoforms

- Example: lactate dehydrogenase (LDH)
  - two types of subunit
    - one predominantly in the heart
    - one predominantly in skeletal muscle
    - LDH is a tetramer, therefore 5 different combinations possible
- Increase in any LDH form in the blood suggests tissue damage
  - increase in heart LDH suggests a heart attack



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Isozymes – clinical use

- Relative amounts of isozymes in blood are useful for diagnostic purposes
- Creatine kinase (CK) is a dimeric protein which binds to the muscle sarcomere
  - M form is produced in skeletal muscle (MM)
  - B form is produced in brain (BB)
  - Heart produces both types, forms a heterodimer (MB)
  - appearance of brain type in blood suggests stroke or tumour
  - appearance of heart type suggests heart attack
- More than 20 enzymes are typically used in the clinical lab to diagnose disease

Iso-enzymes : these are the multi-molecular forms of the same enzyme, found in the same species, but in different tissues in different amounts .

Examples

1-The four (1,2,3,4) iso-enzymes of hexokinase .

2-The two (1,2) iso-enzymes of malate dehydrogenase .

3-The three (CPK1 =BB, CPK2=MB,CPK3 =MM) of creatine kinase .

4-The five iso-enzymes of LDH :

LDH1	LDH2	LDH3	LDH4	LDH5
H4	H3M	H2M2	H1M3	M4

LDH is a polypeptide tetramer made up from two different polypeptide monomers : H (heart) and M (muscle), so there are five possible tetramers.

CLINICAL ASPECTS OF ENZYMOLOGY

Now a days 25% of the work of an ordinary clinical chemistry laboratory consists of enzyme activity estimation, and up to about (20) different enzymes are assayed routinely.

Usually, serum is the fluid in which most enzyme measurement are made. Urine, RBC, WBC, CSF, biopsy materials and cultures may be used in certain cases.

Enzymes present in human body can be considered in of the two types :

1-Plasma specific`(functional) enzymes:

These are enzymes which are present Continuously in the plasma, their substrates are also present in plasma, so these enzymes perform physiologic functions in blood.

Examples:

Enzymes concerned with blood coagulation, lipoprotein metabolism enzymes.

2-Plasma non-specific (non-functional) enzymes = intra-cellular enzymes:

These are enzymes which are present under normal condition only in traces in plasma (normal destruction of cells), their substrates are frequently not present in plasma, so they perform no physiologic function in plasma.

The presence of these enzymes in abnormal amounts in serum or plasma indicates tissue destruction, tissue necrosis, tissue inflammation or change of permeability .

Examples : sGPT, sGOT, ACP, ALP---------etc.

The principal advantages of enzymology are :

1-As a mean of diagnosis .

2-As a mean of therapy .

5-As a mean for determination of certain metabolites of clinical importance .

1-As a mean of diagnosis :

The determination of enzyme activity can help the physician to confirm the suggestive diagnosis:

1-CPK activity increases in cases of skeletal and heart diseases .

2-The activity of alpha amylase increases in acute and chronic pancreatitis while decrease in liver disease.

3-The activity of alkaline phosphatase increases in bone diseases.

4-The activity of acid phosphatase increases in prostatic carcinoma.

5-The activity of LDH increases in heart diseases.

6-The activity of AST increases in hepatic diseases.

Organ enzyme tests (enzyme biomarkers)

1.Liver

a. **Serum enzymes that reflect damage to hepatocytes**

i. **Aspartate transaminase (AST)**

ii. **Alanine transaminase (ALT)**

b. Serum enzymes that reflect cholestasis or biliary obstruction:

i. **Alkaline phosphatase (ALP)**

ii. **Gamma glutamyl transpeptidase (GGT)**

iii. **5′-nucleotidase**

Pancreas -2

a.Alpha amylase

b.Pancreatic lipase

Heart (cardiac enzymes) -3

a.Creatine kinase (CK-MB)

b.Aspartate transaminase (AST), or glutamate oxalate transferase (GOT).

c.Lactate Dehydrogenase (LDH)(LDH-H4)

Other enzymes which have been studded but not commonly done:

a.Gamma glutamyl transferase (GGT)

b.Histaminase

c.Pseudocholinesterase

Bones -4

a.Alkaline phosphatase (ALP)

Prostate -5

a.Acid phosphatase (ACP)

Skeletal muscles -6

a.Creatine kinase (CK3)(CK-MM)

b.Lactate dehydrogenase (LDH-M4)

2-As a mean of therapy :

1-The removal of toxic substances from the blood.

2-Genetic deficiency diseases.

3-The enzyme asparaginase is used in the treatment of certain type of leukaemia.

4-Degradation of necrotic tissues by use of proteolytic enzymes such as trypsin and chemotrypsin .

5-Removal of blood clots using streptokinase or urokinase.

6-Treatment of pancreatic insufficiency which occurs in cystic fibrosis by administration of suitably entrapped enzymes such as proteolytic enzymes.

3-As a mean for determination of clinically important metabolites :

a-Determination of blood glucose using glucose oxidase and peroxidase.

b-Determination of serum urea using urease.

c-Determination of serum uric acid using urikase enzyme.